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Reflection Journal

Going into this lab, I thought the hardest part of Natural Language Processing was dealing with high amounts of information or dealing with multiple processes that I had to run and know. What actually happened was the errors I came through in google collaborate, having little knowledge on python I was stuck trying to figure out what was going wrong, but I found out the real reason was every variable downloaded only stick around while the session was alive, so if the system restarts nothing is saved I had to run everything from top to bottom. This helped solve my issue in this lab and gave me an insight that I need to look everywhere to solve an issue.

After finishing the lab, it clicked for me thinking about search engines. How if I search for the best place, the engine would narrow it down to what I'm looking for. This gave me a deeper insight on how this connects to day to day usage. It made me think more deeply about chatbots and translation apps, and I realized these functions as a cleanup step before handling the work they were asked to do. I thought a lot about the choices I made in this lab. Not stripping out a word could change the course of the performance.

NLTK and spaCy are two things that felt pretty different. At the start, NLTK wanted me to download things separately to make sure it could work. It was like building my toolbox and adding more things to it one by one. We needed the NLTK implementation for spaCy to work, so it was built in. What I got from this was that NLTK can be used step by step, while spaCy is better once you already know what is going on.

Both stemming and lemmatization claim to reduce a word to its root, yet they offer options that offset them. Stemming offers a quicker method to reduce a word, but it can misspell words. Chatbots can use stemming because it provides a faster response . While for lemmatization it used grammar to land on a proper word for the chopping, but the negative of this it takes long for it to happen, having an outcome like that both options weigh each other out. Both tradeoffs of stemming used for speed at scale while lemmatization meaning actually matters. There are many other things I can talk about, such as stop word removal, text cleaning, and additional implements, but I'm going over the big takeaways.

From this lab, it gave me a big insight on how preprocessing is the first step in a project, not a step to rush through, but to classify a sentiment. I like to lean towards spaCy and

lemmatization since it can give me a deeper insight into the word meaning matters, and will be more cautious on stripping words. The NLTK approach or even stemming seems good to use since exact meaning matters less than speed. But after this lab, it gave me an insight into the process of NLP and a habit of tracing an error back, I will be carrying this into every project from now on.